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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations
May 2025

CS2501-1- FUNDAMENTALS OF AI AND ML
(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Who is considered the father of Artificial Intelligence?**
A) Alan Turing
B) John McCarthy
C) Marvin Minsky
D) Claude Shannon
2. **Which mathematical concept is the foundation of AI?**
A) Trigonometry
B) Geometry
C) Logic and probability
D) Abstract Algebra
3. **Which AI system was an early example of a "chatterbot"?**
A) ELIZA
B) IBM Watson
C) Siri
D) ChatGPT
4. **What is the main focus of AI research today?**
A) Rule-based systems
B) Machine learning and deep learning
C) Chess-playing programs
D) Symbolic logic
5. **What is Subjective Intelligence?**
A) Fixed intelligence measure
B) Intelligence based on genetics only
C) Intelligence calculated only using IQ
D) Intelligence that depends on perspective and context
6. **How is AI intelligence often compared to human intelligence?**
A) By evaluating problem-solving and learning abilities
B) By testing emotions
C) By analyzing human DNA
D) By measuring muscle reflexes
7. **What is one major challenge AI faces in the future?**
A) Lack of applications
B) Ethical concerns and bias
C) Reducing internet usage
D) Decreasing need for AI in industries
8. **Which AI concept helps machines make decisions with uncertain data?**
A) Fuzzy Logic
B) Hard Coding
C) Fixed Programming
D) Symbolic AI
9. **In an Expert System, what is the main purpose of conflict resolution?**
A) To update the knowledge base automatically
B) To choose among multiple applicable rules when more than one is triggered
C) To prioritize user inputs over system rules
D) To generate new rules based on past errors

10. **Forward chaining in an Expert System can be described as:**
 A) Starting with a goal and working backward to check if data supports it
 B) Applying rules only when triggered by user input
 C) Starting with available facts and using them to infer new facts until a conclusion is reached
 D) Randomly selecting rules to apply
11. **Expert Systems are particularly well-suited for:**
 A) Dynamic learning in unstructured environments
 B) Real-time adaptation and self-improvement
 C) Large-scale data mining with minimal human intervention
 D) Domains where knowledge can be clearly articulated as rules
12. **In backward chaining, the inference process is initiated by:**
 A) The set of available facts
 B) Random selection of rules
 C) The hypothesis or goal that needs to be verified
 D) Continuous rule updates
13. **Fuzzy Logic differs from traditional Boolean logic by:**
 A) Using only binary true/false values
 B) Allowing values to be expressed as degrees of truth
 C) Eliminating the need for if-then rules
 D) Replacing all symbolic reasoning with statistical methods
14. **The overall goal of problem solving in Classical AI systems is to:**
 A) Mimic human logical reasoning and decision-making using symbolic methods
 B) Perform statistical data analysis
 C) Replace human experts completely with machine learning
 D) Generate random outputs for creativity
15. **What is the key advantage of incorporating fuzzy logic into expert systems?**
 A) It converts all inputs into binary values
 B) It eliminates the need for rule-based reasoning
 C) It enables reasoning under uncertainty by allowing partial truth values
 D) It reduces computational requirements significantly
16. **In a system using multiple rules, conflict resolution ensures that:**
 A) All rules are applied simultaneously
 B) Only the simplest rule is used
 C) Rules are applied in alphabetical order
 D) The most specific or highest-priority rule is selected when rules conflict
17. **Supervised learning requires**
 A) No labels for training data
 B) Labeled input and output data
 C) Random selection of inputs
 D) Reinforcement from an agent
18. **Unsupervised learning is mainly used for:**
 A) Predicting stock prices
 B) Labeling new training data
 C) Discovering hidden patterns in data
 D) Teaching robots how to walk
19. **Why is it important to split data into training and testing sets?**
 A) To improve model interpretability
 B) To evaluate the model's ability to generalize to unseen data
 C) To ensure faster computations
 D) To reduce computational storage
20. **Which metric is commonly used for classification problems?**
 A) Mean Square Error
 B) R-Squared
 C) Root Mean Square Error
 D) Precision and Recall

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Artificial Intelligence (AI) and explain its foundational principles.	8	L*1	1	1
	c) Discuss the significance of Alan Turing's contributions to AI.	8	L2	2	1

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|-------|--|---|----|---|---|
| 2. a) | Compare the historical phases of AI development from its early history to its renaissance, | 6 | L2 | 1 | 1 |
| b) | Summarize the role of McCulloch and Pitts in AI Development. | 4 | L2 | 1 | 1 |
| c) | Why Comparative Intelligence is Crucial for AI Development? Explain. | 6 | L2 | 2 | 1 |
| 3. a) | Summarize how neural networks contributed to AI's resurgence in the late 20th century. | 8 | L1 | 2 | 1 |
| b) | Explain technological advancements in the 21st century helped AI grow rapidly. | 8 | L2 | 1 | 1 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | What is an expert system, and what are the key components that make it work? How does forward chaining help in the decision-making process? | 8 | L1 | 3 | 1 |
| c) | How does fuzzy logic differ from traditional Boolean logic in AI? Explain with an example. | 8 | L2 | 4 | 2 |
| 5. a) | What are the key challenges AI faces in problem-solving tasks? | 6 | L1 | 3 | 1 |
| b) | What is defuzzification and how does it impact decision-making? Illustrate. | 4 | L2 | 4 | 2 |
| c) | Explain how AI systems solve complex problems like route optimization with underlying algorithms. | 6 | L2 | 4 | 1 |
| 6. a) | How does an expert system handle multiple rules to reach a conclusion? | 8 | L2 | 3 | 1 |
| b) | What is a fuzzy expert system? How do fuzzy expert systems improve upon traditional expert systems? Explain. | 8 | L2 | 4 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What is Machine Learning and what are its fundamental components? How do "Error" and "Noise" affect machine learning models and what are some strategies to mitigate their impact on model performance? | 8 | L1 | 5 | 2 |
| b) | What is supervised learning? Explain any three common supervised learning algorithms in detail. | 8 | L2 | 5 | 1 |
| 8. a) | What is Unsupervised Learning and How Does it Differ from Supervised Learning? Explain any two commonly used Unsupervised Learning algorithms. | 8 | L2 | 5 | 1 |
| b) | Explain any four real-world applications of machine learning in detail. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
